Multilingual Technologies: An Interdisciplinary Master's Program Leveraging Technology for Language

Sigrid Schefer-Wenzl
Computer Science
University of Applied Sciences Campus Vienna
Vienna, Austria
sigrid.schefer-wenzl@fh-campuswien.ac.at

Igor Miladinovic
Computer Science
University of Applied Sciences Campus Vienna
Vienna, Austria
igor.miladinovic@fh-campuswien.ac.at

Dagmar Gromann
Centre for Translation Studies
University of Vienna
Vienna, Austria
dagmar.gromann@univie.ac.at

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***Abstract*—In today's globalized and increasingly technological world, skills in transcultural communication and multilingual language technology development—ranging from speech recognition to machine translation—are qualifications highly demanded in the job market. The interdisciplinary field of language technology and computational linguistics equips university students with these essential skills. However, in Austria a study program focusing on this interdisciplinary field was missing. The Master's program *Multilingual Technologies*, now offered jointly by the University of Vienna and the University of Applied Sciences Campus Vienna, aims to bridge the gap between language and technology while fostering expertise in multilingual applications. This paper explores the program's interdisciplinary approach, highlighting its design, collaborative framework, and its implications for both academia and the job market. We also present first students' evaluations results, which indicate a high satisfaction among students and possible room for future improvements.**

Keywords—higher education, interdisciplinary curriculum, multilingual technologies, computer science education

I. Introduction

The demand for multilingual communication technologies has grown rapidly in recent years, driven by advancements in neural networks and natural language processing (NLP) technologies like ChatGPT [1, 2]. As global businesses, organizations, and societies become more interconnected, the ability to communicate across languages and cultures is critical. NLP technologies are increasingly employed in applications ranging from automated customer service and cross-border e-commerce to healthcare and public administration [3]. However, expertise in developing and improving these technologies, particularly in multilingual and cultural contexts, remains limited. This gap is especially evident in the predominance of English and a few other major languages in NLP research and development, which leads to challenges in adapting models for low-resource languages [4]. Multilingual language models such as mBERT and XLM-R have shown progress, but they still struggle with consistent performance across diverse linguistic structures, presenting an ongoing challenge in developing robust multilingual NLP systems [5].

Recognizing these needs, the Multilingual Technologies program was launched in 2022 as a joint initiative by the University of Vienna and the University of Applied Sciences Campus Vienna. This program is designed to offer an interdisciplinary master's education that bridges the fields of translation studies, linguistics, and computer science, equipping graduates with skills not only in technical NLP but also in culturally and linguistically informed approaches to language technology. Furthermore, the program fosters a collaborative learning environment that includes hands-on projects with real-world applications, such as developing automated translation tools and cross-linguistic sentiment analysis models.

This paper outlines the structure, collaboration, and innovative aspects of the Multilingual Technologies program, underscoring its contributions to academic and professional domains. The program's unique focus on multilingual and culturally nuanced NLP responds to the industry need to move beyond a monolingual, English-centric paradigm and address the digital linguistic divide. Moreover, it prepares students for emerging challenges in ethical AI, where issues like fairness and representation in language technologies are increasingly critical [6]. By training students to develop inclusive, context-sensitive NLP systems, the Multilingual Technologies program is positioned to make significant contributions to the fields of computational linguistics, translation studies, and AI ethics.

This paper is structured as follows. Section 2 provides an overview of the Multilingual Technologies master's program. Section 3 compares the program to other similar programs offered in Europe. Section 4 introduces evaluation results from the first two cohorts. Section 5 concludes the paper.

II. Description of The Master's Program

The Master's in Multilingual Technologies is an interdisciplinary program jointly offered by the University of Vienna and the University of Applied Sciences Campus Vienna. This program focuses on equipping students with technical, linguistic, and cultural skills necessary for developing and deploying multilingual communication technologies. The curriculum is designed to meet the growing need for experts who can work at the intersection of linguistics, translation studies, and computer science, particularly in fields like natural language processing (NLP), machine translation, and cross-linguistic digital communication.

A. *Curriculum and Structure*

The program is structured into four semesters (two years) and includes a combination of coursework, practical projects, and a final thesis. The curriculum is divided into several key modules that ensure a comprehensive education in both theoretical and applied aspects of multilingual language technologies:

1. Core Modules in Multilingual Technologies: These foundational courses introduce students to the basics of computational linguistics, machine translation, and multilingual NLP. Topics include syntax, semantics, and language processing, emphasizing multilingual applications and data-driven approaches in diverse linguistic contexts.
2. Translation and Localization Technologies: These courses cover translation memory systems, terminology management, and localization processes. They teach students how to leverage these tools to enhance translation workflows, enabling them to build efficient, culturally sensitive multilingual content pipelines.
3. Machine Learning and NLP: Students take advanced courses in machine learning, deep learning, and natural language processing, including areas like sentiment analysis, multilingual data mining, and text classification. These courses provide the technical foundation necessary to work with large multilingual datasets and apply machine learning models in a language technology context.
4. Software Engineering and Human Computer Interaction: The advanced courses in the area of software engineering are specifically tailored to the needs of multilingual software applications. Students learn best practices on how to design and develop such applications applying state of the art user experience guidelines.
5. Cross-Cultural Communication and Ethics in AI: Recognizing the ethical dimensions of multilingual technologies, these courses address the impact of AI on different language communities and cultures. Courses cover the ethical and societal implications of AI, preparing students to consider fairness, representation, and inclusion in their designs.
6. Hands-on Practicum and Internship: Students engage in practical training through internships and project-based courses. This hands-on experience is critical, as it allows students to apply their skills in real-world contexts, often collaborating with industry partners or working on applied research projects.
7. Master's Thesis: In the final semester, students conduct original research on a topic related to multilingual technologies. The thesis project allows students to synthesize their learning and contribute new insights to the field, often focusing on real-world challenges in multilingual technology.

Collaboration between the University of Vienna and University of Applied Sciences Campus Vienna is a hallmark of the Multilingual Technologies program. The entire student journey—from admission to final examinations—is jointly managed by both institutions. Students have access to the combined resources of both universities and can benefit from the expertise of instructors in both fields. Thesis supervision is similarly collaborative, with students choosing a supervisor from either institution depending on their research focus.

One of the program's defining features is its strong emphasis on interdisciplinarity. The curriculum was co-developed by faculty members from both institutions, combining the University of Vienna's strengths in linguistics, translation, and transcultural studies with the University of Applied Sciences Campus Vienna's expertise in computer science and software engineering. The curriculum includes modules that bridge both fields, such as "Multilingual and Cross-lingual Methods and Language Resources" and "Human-Computer Interaction for Computational Linguists". Additionally, courses such as statistics and transcultural communication focus on providing the theoretical foundation necessary for research in multilingual technologies.

This interdisciplinary approach allows students to engage with both foundational research and practical applications, thereby equipping them with the ability to develop cutting-edge multilingual technologies that are sensitive to diverse linguistic and cultural contexts. This aligns with previous research that shows interdisciplinary curricula are effective in fostering comprehensive, problem-solving skills in emerging fields [7].

B. *Learning Goals*

The Multilingual Technologies program aims to prepare students with a mix of technical, linguistic, and cultural competencies that enable them to work effectively in multilingual and cross-cultural contexts. The primary learning goals include:

Technical Proficiency: Graduates gain a strong foundation in computational linguistics and machine learning, enabling them to build, evaluate, and improve multilingual NLP models and applications.

Interdisciplinary Knowledge: The program fosters an interdisciplinary understanding, blending insights from linguistics, translation studies, and computer science. This equips students with the ability to design language technology solutions that are both technically sound and culturally informed.

Multilingual Data Management Skills: Students learn to manage multilingual datasets, work with under-resourced languages, and use translation technologies, essential for building and refining language models across different languages.

Ethical and Cultural Awareness: The program emphasizes the importance of ethical considerations in multilingual technology, teaching students to approach technology development with sensitivity to cultural differences and social impact.

Practical Application and Project Management: Through internships and real-world projects, students gain hands-on experience, preparing them to manage language technology projects from conception through deployment in industry settings [8].

C. *Preparatory Extension Curricula*

Each year, 30 students are admitted, with applicants coming from diverse backgrounds, typically with undergraduate degrees in either translation studies or computer science. Due to the interdisciplinary nature of the program, students are required to acquire foundational knowledge in the complementary discipline prior to starting the master's program. To facilitate this, the University of Vienna offers an extension curriculum in "Specialized Communication and Language Technologies" (15 ECTS), while the University of Applied Sciences Campus Vienna provides an extension curriculum in "Computer Science" (15 ECTS).

The contents of these extension curricula serve as admission criteria, ensuring that students enter the master's program with a solid understanding of both fields. By building this common knowledge base, the program seeks to prevent potential dropouts and ensure that students are prepared for the interdisciplinary challenges ahead. This approach aligns with educational best practices for interdisciplinary programs [9], which emphasize the need for pre-entry bridging courses in complementary fields to foster successful interdisciplinary learning.

D. *Master's Theses Topics*

Topics of master's theses range from multilingual detection of explicit and implicit hate speech, impact of code-switched rap data on neural language model performance to automated speech recognition in the German medical domain and preserving privacy in information extraction. One highly innovative master's thesis focused on a linear transformation of hidden representations of language models in order to identify the models' ability to learn factual associations only from text. It turns out that this type of transformation results in vector spaces better equipped for knowledge-intense tasks. In NLP it is common to evaluate approaches on standard benchmarks, which provides little information on their transferability to practical settings. To this end, a master's thesis on text summarization explored the applicability of the most successful neural methods to the real-world setting of research on the credibility of asylum claims, providing expert evaluations on the automatically generated summaries. Even though human evaluations are still required, the summary generation considerably accelerates the process of asylum research and provided excellent starting points.

Hate speech detection is an important topic as it is impossible to humanly moderate online contents due to the sheer amount of text. Thus, several master's theses approach this topic from different perspectives. In hate speech, it is generally common to mix languages, e.g. use swear words in English and German within the same online post. Thus, one thesis explored the improved ability of neural approaches to detect code-mixed hate speech by training on both languages involved in the mixing. Humans tend to not always agree whether something is hate speech or not. This was explored by another master's thesis by explicitly incorporating disagreement of human annotators in the model training process in Italian, Slovenian and English. Finally, hate speech does not necessarily require the use of swear words but can take a highly implicit form, which makes it harder to detect automatically. To address this issue, one master's thesis explored the ability of large language models to directly detect implicit hate speech or identify specific features of such speech that is then used in traditional classifiers.

More information about master's thesis topics can be provided on request. Please contact the authors of this article.

E. *Limitations of the Program*

While the Multilingual Technologies Master's program provides a robust interdisciplinary foundation, it has some notable limitations. One key challenge is its broad curriculum, which, while offering a diverse array of topics, may limit the depth of specialization. Students must divide their focus between computational linguistics, translation technology, and computer science related topics, which can make it difficult to gain expert-level knowledge in a single or combined area. For those seeking advanced technical training specifically in machine learning or deep learning for NLP, for example, this program may not provide the depth found in more computationally intensive master's programs. Additionally, while the program includes practical training, the hands-on experience with the latest NLP tools and frameworks may be constrained by the need to cover a wide range of skills, leaving students with a generalist approach rather than deep technical expertise.

Another limitation of this program lies in the administrative complexity that arises from the collaboration between the University of Vienna and the University of Applied Sciences Campus Vienna. Students enrolled in the program must often navigate the logistical challenges of coordinating between two separate institutions, having different administrative systems, academic calendars, and program requirements. For instance, registering for courses, accessing campus resources, or applying for financial support requires students to deal with two distinct sets of administrative protocols. This creates an added layer of complexity for students compared to a more integrated, streamlined experience typical of single-institution programs.

Additionally, the dual-institution setup can lead to scheduling conflicts and a sense of divided academic identity. Students have to travel between campuses for different classes or face challenges in accessing faculty support and academic advising due to the physical separation between institutions. Furthermore, coordinating joint events, workshops, or guest lectures across two institutions can be logistically challenging.

Finally, each year we are facing a growing number of applicants which are not able to fulfil admission criteria. This is mainly because we require students to have foundational knowledge in two very different domains. We addressed this issue by offering extension curricula as stated in Section II.C. However, for students not (yet) based in Vienna, this is not a feasible option.

III. COMPARISON TO SIMILAR PROGRAMS

There is no comparable program to Multilingual Technologies in Austria. However, in Europe there are several similar programs, for example:

- The University of Stuttgart offers a Master's in Computational Linguistics that covers statistical and machine-learning approaches to language processing. The program also includes specialized courses in semantics, syntax, and phonetics, along with NLP applications.
- The University of Amsterdam offers a Master's in Artificial Intelligence with specializations in Natural Language Processing and Computational Linguistics. The program is highly regarded for its focus on deep learning, language modelling, and machine translation.

- The University of Edinburgh provides a Master's in Speech and Language Processing, a highly competitive program covering topics in NLP, linguistics, and deep learning.
- The Master's in Machine Learning and Machine Intelligence at the University of Cambridge includes modules in natural language processing and computational linguistics. It focuses on mathematical and computational theory.
- The Master in Language Science and Technology at Université Paris-Saclay offers a strong foundation in computational methods applied to linguistics.

The Master's in Multilingual Technologies at the University of Vienna is distinct in several ways from traditional computational linguistics and NLP-focused programs, as it places an emphasis on multilingualism, interdisciplinary practical applications, and human-centred approaches to language technology (see also Table 1).

Unlike many study programs that emphasize monolingual language processing (often English-centric), our Multilingual Technologies program is designed to tackle the challenges of multilingual NLP and cross-linguistic processing. This includes machine translation, localization, and working with diverse language resources. Students gain hands-on experience with translation memory systems, terminological databases, and other tools that support multilingual content management.

Students learn to work with tools and methods that support multiple languages, making it particularly relevant for industries requiring multilingual applications, such as international business, translation services, and multinational corporations.

Furthermore, the program incorporates a strong human-centred approach, focusing on the societal and ethical aspects of language technology. It prepares students to consider how language technologies impact society and users from different cultural and linguistic backgrounds. Unlike some computational linguistics programs that primarily focus on algorithmic and technical training, Vienna's program encourages students to understand the broader context and ethical implications of multilingual technologies, preparing them for roles where *user-centred design* and *ethical deployment* are important.

TABLE I. KEY DIFFERENCES TO SIMILAR PROGRAMS

Key differences	
Multilingual Technologies	***Other programs***
Emphasis on multilingual NLP and translation technology	Often focused on monolingual (primarily English) NLP
Human-centered and ethical approaches to tech design	More focus on technical and algorithmic development
Interdisciplinary (linguistics, translation, digital humanities, computer science, software engineering)	Often computer science-driven with some linguistics
Career paths in translation, localization, multilingual data management	Career paths in NLP, AI, computational linguistics research

Our Multilingual Technologies program draws from linguistics, translation studies, and computer science, creating a unique interdisciplinary environment. While other programs may be housed entirely within a computer science department, the Vienna program is structured to include insights from linguistic theory, translation studies, and digital humanities. This interdisciplinary setup allows students to gain expertise in both the technical aspects of computational linguistics and the theoretical underpinnings of language use across cultures.

IV. Evaluation Results

The Multilingual Technologies master's program has received positive evaluations from students, who appreciate the rich, interdisciplinary curriculum. Many students express satisfaction with the depth and relevance of the course content, which equips them with a blend of technical and linguistic skills highly applicable to multilingual and cross-cultural technology fields. Students also value the quality and expertise of the lecturers, who bring a wealth of knowledge from diverse research and industry backgrounds, creating a well-rounded learning environment. The lecturers' commitment to teaching and their real-world experience help students build a strong, applicable skill set, and students often mention that the course materials and projects reflect current industry trends and challenges in multilingual technology.

In the following we will show a selection of questions of the survey among the first cohort, which has finished their study in September 2024. Ten students participated in this survey and answered questions using a scale between 1 and 6, whereas 1 means "fully disagree" and 6 means "fully agree".

Fig. 1 shows the overall satisfaction with the study program. The average rating of 4.7 shows relatively high satisfaction, but also some room for improvement (see Section II.E).

Fig. 1. Student answers to the question on overall satisfaction

In Fig. 2 we can see how students evaluate their technical skills development during the study. With the average rating of 5.75 students believe that with the study program they develop relevant technical skills. We have got the same rating on questions related to the development of professional skills, such as teamwork, problem solving and organisational skills.

Finally, Fig. 3 depicts students' self-evaluation on the ability to apply their theoretical knowledge in practical projects. Also here, with the average rating of 5 students are confident that they are able to contribute with their obtained skills to practical projects.

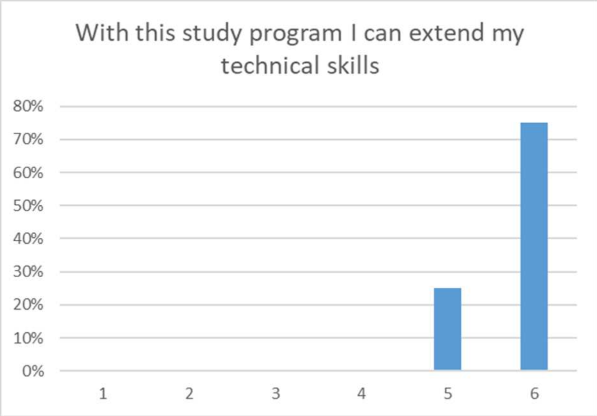

Fig. 2. Student answers to the question on skills development

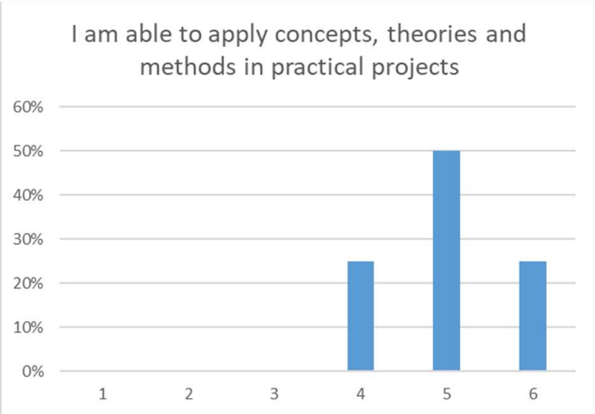

Fig. 3. Student answers to the question on practical skills

However, despite the academic strengths of the program, students have reported frustrations with certain administrative processes, largely stemming from the coordination between two different institutions. Managing administrative tasks across two institutions has been a recurring pain point, with students noting delays and difficulties in registration, accessing resources, and obtaining consistent support. Differences in each institution's administrative procedures and timelines can lead to confusion, especially when it comes to course enrolment, schedule management, and communication between departments. Addressing these administrative challenges remains an area for improvement to enhance the overall student satisfaction and streamline the educational experience.

V. Conclusion

The Multilingual Technologies master's program represents an innovative, interdisciplinary approach to language and technology education. Its collaborative framework, combining fundamental research with applied skills, ensures that students are well-prepared for careers in a rapidly evolving field. The program's strong emphasis on multilingualism, practical experience, and interdisciplinary collaboration positions it as a leading initiative in the field of language technologies.

As the first cohort has just completed the program, it is clear that the program is addressing a significant gap in both academia and the job market. With increasing demand for experts in multilingual language technologies, this program is well-placed to contribute valuable knowledge and skills to the workforce and society. Furthermore, the success of this initiative highlights the importance of cross-institutional collaboration in addressing emerging challenges in higher education and industry.

In our future work, we plan to offer an intensive study preparation program to help more applicants to fulfil admission criteria. This program should be accessible for applicants missing the knowledge in one of the two disciplines.

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